

Experiment 10

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AIM: The aim is to create data visualizations, such as line charts and doughnut charts, for a student performance analysis system using JavaScript and Chart.js.

PROCEDURE:

Step 1: Set Up Your HTML File Create an HTML file to define the structure, including canvas elements where the charts will be rendered, and link the Chart.js library.

HTML (index2.html):

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Student Performance Charts</title>
```

```
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
```

```
  <style>    body { font-family: Arial, sans-serif; background-color: #f4f4f4; text-align:
center; }    h2 { color: #222; }    canvas {      display: block;      margin: 20px
auto;      background-color: #fff;      border: 1px solid #ddd;      border-radius:
6px;      box-shadow: 0 2px 6px rgba(0, 0, 0, 0.1);
    }
```

```
  </style>
```

```
</head>
```

```
<body>
```

```
  <h2>Student Performance Analysis</h2>
```

```
  <h3>Marks Progress (Line Chart)</h3>
```

```
  <canvas id="lineChart" width="400" height="200"></canvas>
```

<h3>Subject-wise Performance (Doughnut Chart)</h3>

<canvas id="doughnutChart" width="400" height="200"></canvas>

<script src="script1.js"></script>

</body>

</html>

Step 2: Create the JavaScript File for Charts Create a JavaScript file to define the datasets for test marks and subject distribution, and initialize the charts using the Chart.js constructor.

JavaScript (script1.js):

JavaScript

// Line Chart - Marks Progress const lineCtx =

document.getElementById('lineChart').getContext('2d'); const lineChart =

new Chart(lineCtx, { type: 'line', data: { labels: ['Test 1', 'Test 2',
'Test 3', 'Test 4'], datasets: [{ label: 'Marks', data: [65, 75,
70, 85], borderColor: 'blue',

fill: false,

tension: 0.3

}}

},

options: { responsive: true }

});

// Doughnut Chart - Subject Performance const doughnutCtx =

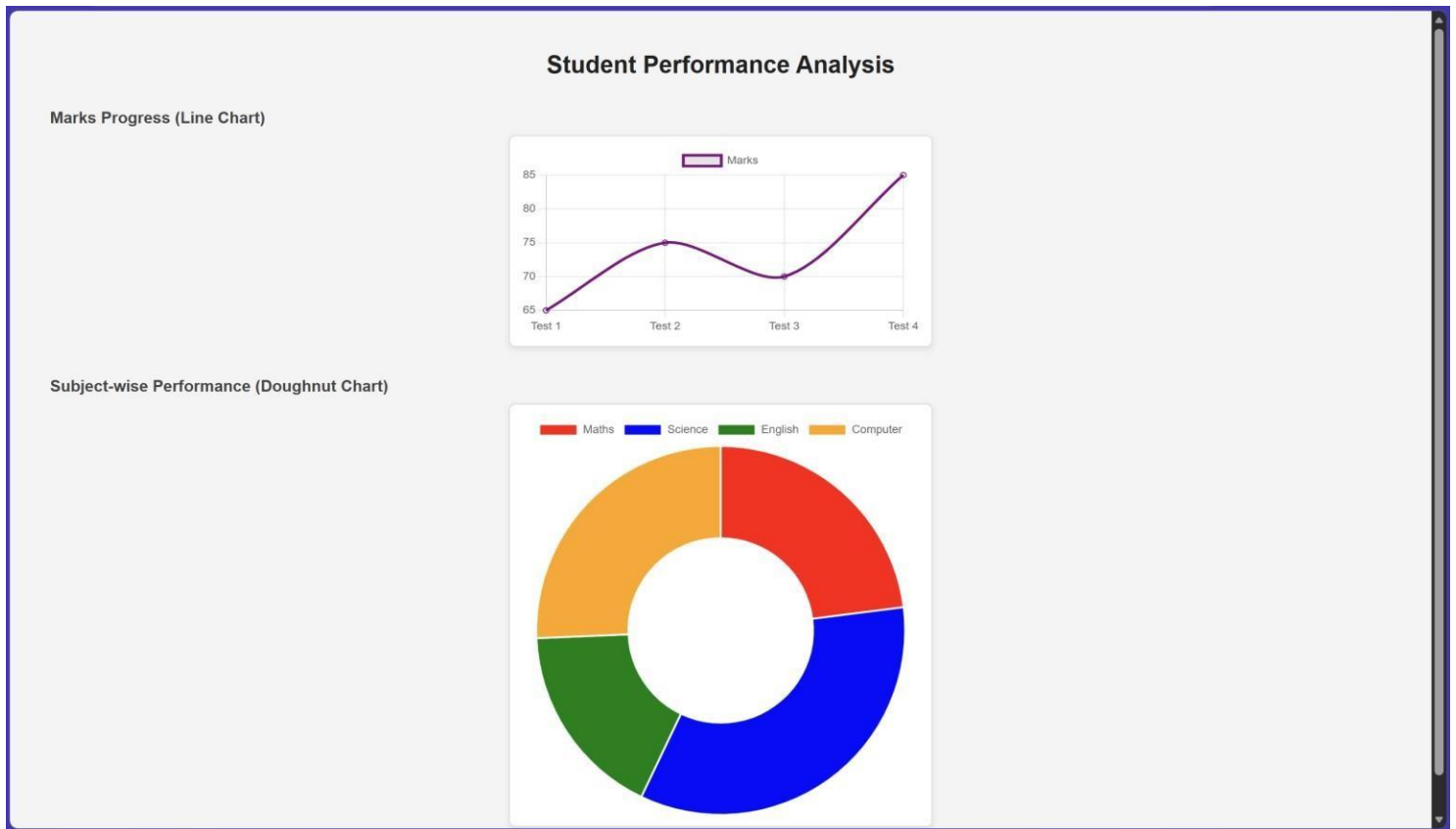
document.getElementById('doughnutChart').getContext('2d'); const doughnutChart =

new Chart(doughnutCtx, { type: 'doughnut', data: { labels: ['Maths', 'Science',
'English', 'Computer'], datasets: [{ label: 'Marks Distribution', data:
[85, 75, 80, 90], backgroundColor: ['red', 'blue', 'green', 'orange']

}}

```
},  
options: { responsive: true }  
});
```

OUTPUT:



RESULT: The data visualizations for student performance were successfully created using JavaScript and Chart.js, providing a clear graphical representation of marks progress and subject-wise distribution.